# week 5

# Chain rule (continued)

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

what if there are more than 2 functions

$$\frac{d}{dx} f(g(h(x))) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Ex $\frac{d}{dx} f(2f(3f(x))) = f'(2f(3f(x))) \cdot 2f'(3f(x)) \cdot 3f'(x)$

wow that's simple

Ex

$$g(x) = \left[f\left(\frac{2}{x}\right)\right]^3$$

$$g'(x) = 3\left[f\left(\frac{2}{x}\right)\right]^2 \cdot f'\left(\frac{2}{x}\right) \cdot \left(-\frac{2}{x^2}\right)$$

# Trig - Dih-rivitives

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Proof later

can be used in calculations

Ex $\lim_{x \to 0} \frac{\sin 3x}{\sin 7x} = \lim_{x \to 0} \frac{\sin 3x}{3x} \cdot \frac{7x}{\sin 7x}$ (1 , 1)

$\lim_{x \to 0} \frac{\sin 3x}{3x} = 1$ , $\lim_{x \to 0} \frac{7x}{\sin 7x} = 1$

=> lim is $\frac{3}{7}$

Derrivatives

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos^2 x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x \quad \text{(exercise)}$$

$$\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = -\frac{-\sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \cdot \tan x$$

Note: the trig function d.n formulas assume the argument is in radians

if $x$ is in degrees what is the angle in radians

$$x \text{ in degrees} \longrightarrow \frac{\pi x}{180} \text{ radians}$$

if $x$ is in degrees,

$$y = \sin\left(\frac{\pi x}{180}\right) \Rightarrow y' = \frac{\pi}{180} \cdot \cos\frac{\pi x}{180}$$

$$y'\Big|_{x=45} = \frac{\pi}{180} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{360}$$

$$y_0 = y(45^\circ) = \sin(45) = \frac{\sqrt{2}}{2}$$

Equation of tangent line:

$$y - \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{360}(x - 45)$$

$$\lim_{x \to 0} x^2 \csc x \cot x$$

$$= \lim_{x \to 0} x^2 \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

$$= \lim_{x \to 0} \left(\frac{x}{\sin x}\right)^2 \cos x$$

($\frac{x}{\sin x} \to 1$, $\cos x \to 1$)

$\Rightarrow$ lim is 1.

$y = f(x)$ differentiable

$y' = f'(x) = \frac{dx}{dy}$ first derivative

Which is a function itself

if $f'$ is differentiable:

$$\frac{d}{dx} \cdot \frac{dy}{dx} = \left(f(x)\right)' = f''(x) = \frac{d^2y}{dx^2}$$

repeat: $y^{(n)}(x) = \frac{d^n y}{dx^n} = f^{(n)}(x)$

$n^{th}$ order derivative of $f(x)$

Ex

$f(x) = 3x^3 - 7x^2 + x$

$f'(x) = 9x^2 - 14x + 1$

$f''(x) = 18x - 14$

$f^{(3)}(x) = 18$

$f^{(4)}(x) = 0$

any dirivitive beyond that is zero

Ex $y = f(x) = \frac{1}{x-3}$

$$f'(x) = -\frac{1}{(x-3)^2}$$

$$f''(x) = \frac{2}{(x-3)^3}$$

$$f^{(3)}(x) = -\frac{2 \cdot 3}{(x-3)^4}$$

$$f^{(4)}(x) = \frac{2 \cdot 3 \cdot 4}{(x-3)^5}$$

$1 \cdot 2 \cdot 3 \cdot 4 = 4!$

$1 \cdot 2 \cdot 3 \cdots n = n!$

$$f^{(n)} = \frac{(-1)^n \cdot n!}{(x-3)^{n+1}}$$

Valid for any integer

Ex

$f(x) = \sin x$

$f'(x) = \cos x$

$f''(x) = -\sin x$

$f^{(3)}(x) = -\cos x$

$f^{(4)}(x) = \sin x$

$n = 4k$ (k integer) $\Rightarrow$ $f^{(n)}(x) = \sin x$

$n = 4k+1$ $\quad f^{(n)}(x) = \cos x$

$n = 4k+2$ $\quad f^{(n)}(x) = -\sin x$

$n = 4k+3$ $\quad f^{(n)}(x) = -\cos x$

Ex

$y = f(x) = \sqrt{1-3x}$ Calculate $f'$, $f''$

$$f'(x) = -\frac{3}{2\sqrt{1-3x}} = -\frac{3}{2}(1-3x)^{-\frac{1}{2}}$$

$$f''(x) = \frac{3}{4}(1-3x)^{-\frac{3}{2}} \cdot (-3) = -\frac{9}{4}(1-3x)^{-\frac{3}{2}} \;//$$

Exercise: Calculate $f^{(3)}(x)$

# Using derivative and differentials

$y = f(x) \qquad y' = \frac{dx}{dy} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$

$dy, dx \rightarrow$ differentials

they are $\Delta y$ and $\Delta x$ as they $\rightarrow 0$

$$dy = y' dx \Rightarrow \boxed{\Delta y \approx y' \cdot \Delta x}$$

Can be used to approximate small changes in the value of a function

**Ex** $f(x) = \sqrt{3x+1}$

Estimated the change in f as x increase from $a=1$ to $x=1.08$

$\Delta y \approx f'(a) \cdot \Delta x$

$f'(x) = \frac{3}{2\sqrt{3x+1}} \qquad f'(1) = \frac{3}{2 \cdot 2} = \frac{3}{4}$

$\Delta y \simeq \frac{3}{4} \cdot 0.08 = 0.06$

Exact value: $f(1.08) = \sqrt{3 \cdot 1.08 + 1} = 2.059126$

$f(1) = \sqrt{4} = 2$

$\Delta y_{exact} = f(1.08) - f(1) = 0.059126$

**Example** ★

$y = f(x) = \sqrt{x}$

if $x$ undergoes a negative change of 2% what would be the negative change in $y$? $\frac{\Delta x}{x}$, $\frac{\Delta y}{y}$ negative change

$\Delta y \approx f'(a) \cdot \Delta x = f'(x) \cdot \Delta x$

$f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow \Delta y \simeq \frac{\Delta x}{2\sqrt{x}}$

$\frac{\Delta y}{y} = \frac{\Delta x}{2\sqrt{x} \cdot \sqrt{x}} = \frac{1}{2}\frac{\Delta x}{x}$ $(x > 0)$

if $\frac{\Delta x}{x} = 0.02 \Rightarrow \frac{\Delta y}{y} = 0.01$ or 1%

**Example** what is the rate of change of a side of cube $x$ with respect to its volume $V$?

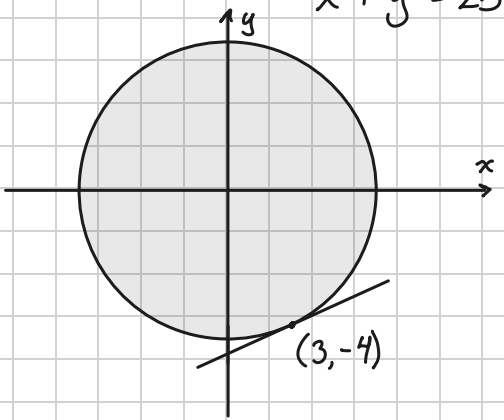
rate of change = derivative

Volume $V$




$V = x^3 \Rightarrow x = f(v) = v^{\frac{1}{3}}$

$\frac{dx}{dy} = \frac{1}{3}V^{-\frac{2}{3}} = \frac{1}{3V^{\frac{2}{3}}}$




find the equation of the tangent line to the circle at $(3, -4)$

$y^2 = 25 - x^2$

$y = \pm\sqrt{25 - x^2}$

$\frac{d}{dx}: \quad x^2 + y^2 = 25$

$2x + 2y \cdot y' = 0 \quad \Rightarrow \quad y' = -\frac{x}{y}$

$y'(3, -4) = -\frac{3}{-4} = \frac{3}{4}$

Equation of tangent line

$y + 4 = \frac{3}{4}(x - 3)$

$y = \frac{3}{4}x - \frac{9}{4} - 4 \Rightarrow$ $\boxed{y = \frac{3}{4}x - \frac{25}{4}}$

**Example** $x^2 + y^2 = -\frac{9}{y^3}$

Find $y'$

$\frac{d}{dx}: \quad x^2 + y^2 + \frac{9}{y^3} = 0$

$2x + 2yy' - 27y^{-4} \cdot y' = 0$

$2x + \left(2y - \frac{27}{y^4}\right)y' = 0$

$y'\left(\frac{27}{y^4} - 2y\right) = 2x \Rightarrow \quad y' = \frac{2x}{\frac{27}{y^4} - 2y}$

$\Rightarrow$ $\boxed{y' = \frac{2xy^4}{27 - 2y^5}}$

Example $x^2 + y^2 = 9$, Find $y'$, $y''$

$\frac{d}{dx}: \quad x^2 + y^2 = 9$

$2x + 2yy' = 0 \quad \Rightarrow \quad y' = -\frac{x}{y}$

$\frac{d}{dx} = x + yy' = 0$

$1 + (y')^2 + yy' = 0$

We already have $y'$: $y' = -\frac{x}{y}$

Sub in here:

$1 + \frac{x^2}{y^2} + yy'' = 0 \quad \Rightarrow \quad yy'' = -\left(1 + \frac{x^2}{y^2}\right)$

$yy'' = -\frac{x^2+y^2}{y^2} \quad \Rightarrow \quad y'' = -\frac{x^2+y^2}{y^3}$

but $x^2 + y^2 = 9 \quad \Rightarrow \quad \boxed{y'' = \frac{9}{y^3}}$

# Antiderivatives

$f(x) = x^2 \qquad f'(x) = 2x$

$2x$ is the derivative of $x^2$

$x^2$ is an antiderivative of $2x$

Def $F(x)$ is an antiderivative of $f(x)$ if $F'(x) = f(x)$ //

$\frac{d}{dx}(x^2) = 2x \Rightarrow x^2$ is an antiderivative of $2x$

$G(x) = x^2 + C$ is also an antiderivative of $2x$

$\frac{d}{dx}(x^2 + C) = 2x$ //

if $F(x)$ is an antiderivative of $f(x)$ then so is $f(x)+c$

Antiderivative = indefinite integral of $f(x)$

$$\int f(x)\,dx = f(x)+c$$

Examples

$$\int 2x\,dx = x^2+c \quad \Rightarrow \int x\,dx = \frac{x^2}{2}+c$$

$$\int 3x^2\,dx = x^3+c \quad \Rightarrow \int x^2\,dx = \frac{x^3}{3}+c$$

$$\int x^a\,dx = \frac{x^{a+1}}{a+1}+c, \qquad a \neq 1$$

$$\int \cos x\,dx = \sin x + c$$

$$\int \sin\,dx = -\cos x + c$$

$$\int \sec^2 x\,dx = \tan x + c$$

Ex

$$\int \left(2x^2 - \frac{1}{\sqrt{x}} + x^{\frac{2}{5}}\right) dx$$

$$= 2\int x^2 dx - \int x^{-\frac{1}{2}} dx + \int x^{\frac{2}{5}} dx$$

$$= 2 \cdot \frac{x^3}{3} - \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + \frac{x^{\frac{7}{5}}}{\frac{7}{5}} + C$$

$$= \frac{2}{3}x^3 - 2x^{\frac{1}{2}} + \frac{2}{5}x^{\frac{7}{5}} + C$$

Ex

$$\int \frac{6(x-1)}{x^{\frac{4}{3}}} dx = 6\int x^{-\frac{1}{3}} dx - 6\int x^{-\frac{4}{3}} dx$$

$$= 6 \cdot \frac{x^{\frac{2}{3}}}{\frac{2}{3}} - 6 \cdot \frac{x^{-\frac{1}{3}}}{-\frac{1}{3}} + C$$

$$= 9x^{\frac{2}{3}} + 18x^{-\frac{1}{3}} + C \;//$$

$C + x^2 = 2x$

$x^2 + 5 = 2x$